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EXAM

Instructions:

- *This is a 2h30 hours exam.*
- ***BE SURE TO PUT THE STICKER ON THE EXAM.***
- *There are a total of 100 points on the test.*
- *You have to answer all the questions.*
- *Use ONLY the space provided for answering the questions.*
- *You must write legibly.*

PART I (5 points)

Consider a production function for which output, y , is produced using labor, l , via the following production function (where $A > 0$ and $f > 0$):

$$y(l) = \max \{0, Al - f\}$$

In this production function, f represents a fixed cost of production.

For $l > f$, does this function exhibit increasing, decreasing, or constant returns to scale? Use the definition of returns to scale to explain your answer.

PART 2 (21 points)

3 points for question 1; 6 points for question 3; 4 points each for questions 2, 4, and 5)

Consider an economy in which output is produced using capital, labor, and intermediate goods:

$$Y_t = \bar{A} (K_t^\alpha L_t^{1-\alpha})^{1-\sigma} (X_t)^\sigma \quad (1)$$

Output is either consumed, saved (to produce capital next period), or used as an intermediate input:

$$Y_t = C_t + I_t + X_t \quad (2)$$

Capital grows according to the following rule:

$$K_{t+1} = (1 - \delta)K_t + I_t \quad (3)$$

The population is constant across time:

$$L_t = \bar{L} \quad \forall t \quad (4)$$

Assume that a constant fraction of output is invested

$$I_t = sY_t, \quad (5)$$

and that a constant fraction of output is used as an intermediate input

$$X_t = rY_t. \quad (6)$$

1 – Use lower case letters to write variables in per capita terms (for example, capital per capita at time t is $\frac{K_t}{L_t} = k_t$). Re-write Equations (1), (2), (3), (5), and (6) in per capita terms.

2 – Use “ss” to denote the steady state value of a particular variable (for example, let k_{ss} be the steady state capital-labor ratio.) Explain why $\delta k_{ss} = sy_{ss}$ (one sentence max). Using this equation, as well as the per-capita version of Equation (6), solve for y_{ss} in terms of $\bar{A}, r, s, \delta, \alpha$, and σ .

3 – Using your answer to question 2, write c_{ss} in terms of $\bar{A}, r, s, \delta, \alpha$, and σ . Then find the r and s which maximize c_{ss} . Use s_{gold} to denote the optimal saving rate, and r^* to denote the optimal intermediate-input utilization rate. For the rest of the problem, assume that $r = r^*$ and $s = s_{gold}$.

4 – Plug your answer from question 3 into y_{ss} to determine the optimal steady-state output-labor ratio solely as a function of δ and the parameters of the production technology ($\bar{A}$, σ and α). Compute the elasticity of per capita GDP with respect to technology, $\varepsilon = \frac{\partial y_{ss}}{\partial \bar{A}} \frac{\bar{A}}{y_{ss}}$.

5 – What, in words, is the interpretation of ε ? (One sentence max.) Is ε large when α is large or small? Is ε large when σ is large or small?

PART III (24 points)
(3 points for each question)

A consumer, who lives for an infinite number of periods, has preferences over two types of goods. The first type of goods (called cash goods) must be purchased with cash, while the second type of goods (called credit goods) can be bought without holding money. The representative agent's preferences are given by:

$$U(c_0, d_0, c_1, d_1, \dots) = \sum_{t=0}^{\infty} \beta^t [\alpha \log c_t + (1 - \alpha) \log d_t] \quad (7)$$

In each period, y_t units of consumption are produced and can be divided among the two goods:

$$y_t = c_t + d_t \quad (8)$$

In Equations (7) and (8), c_t is the amount of the cash good that the consumer purchases in period t , while d_t is the amount of the credit good that he purchases in period t . Assume that y_t is an endowment that grows at rate

g per period:

$$y_t = y_0 \cdot (1 + g)^t \quad (9)$$

The consumer enters period t with m_t dollars in cash, $b_t(1 + R_t)$ dollars in bonds left over from the previous period (i.e., b_t are the bonds purchased in period $t - 1$ that mature in period t). Assume also, that the consumer receives τ_t dollars, as a transfer from the government. In the same period, it is endowed with $P_t y_t$ dollars of the consumption good.

In period t , the consumer pays $P_t(c_t + d_t)$ to buy the cash good and the credit good. It pays b_{t+1} dollars to buy bonds for the next period. Finally, it keeps m_{t+1} dollars in cash.

Putting the last two paragraphs together, the budget constraint of the consumer is:

$$m_t + b_t(1 + R_t) + P_t y_t + \tau_t = P_t(c_t + d_t) + m_{t+1} + b_{t+1} \quad (10)$$

In Equation (10), R_t is the nominal interest rate for bonds that mature in period t .

The cash-in-advance constraint states that the consumer must have enough money to purchase the cash good:

$$P_t c_t = m_t \quad (11)$$

1 – Write the Lagrangian for the constrained maximization problem of the consumer. Use λ_t for the Lagrange multiplier on the period- t budget constraint. Use μ_t for the Lagrange multiplier on the period- t cash-in-advance constraint.

2 – Compute the first order conditions with respect to c_t, d_t, b_{t+1} , and m_{t+1} .

3 – Using Equation (8) plus the four first order conditions, solve for c_t and d_t in terms of y_t , $(1 + R_t)$, and α .

4 – For the moment, assume that $R_t \equiv R \geq 0$ is the same for all periods. Using your answer to question 3, argue why c_t grows at rate g per period. Also, using your answer to question 3, argue why d_t grows at a rate g per period. Finally, solve for P_t as a function of y_t , m_t , α , and R .

5 – Using your first order conditions, and the fact that c_t (also d_t) grows at a constant rate across periods, find an expression relating $\frac{P_{t+1}}{P_t}$, β , g , and $(1 + R)$.

6 – Suppose $\tau_t = \gamma m_t$, so that the per capita money supply grows at a rate γ per period (γ may be different from g). Compute the inflation rate $\pi_t = \frac{P_{t+1}}{P_t} - 1$. (Hint: Use the cash-in-advance constraint— $P_t c_t = m_t$ —for periods t and $t + 1$.)

7 – Using your answers to earlier questions of the problem, solve for c_t in terms of α, β, γ , and y_t . Then solve for d_t as a function of α, β, γ , and y_t .

8 – What is $\frac{\partial c_t}{\partial \gamma}$? What is $\frac{\partial d_t}{\partial \gamma}$? What is the interpretation of these results? (2 sentences max)

PART IV *50 points*
(*10 points for each question*)

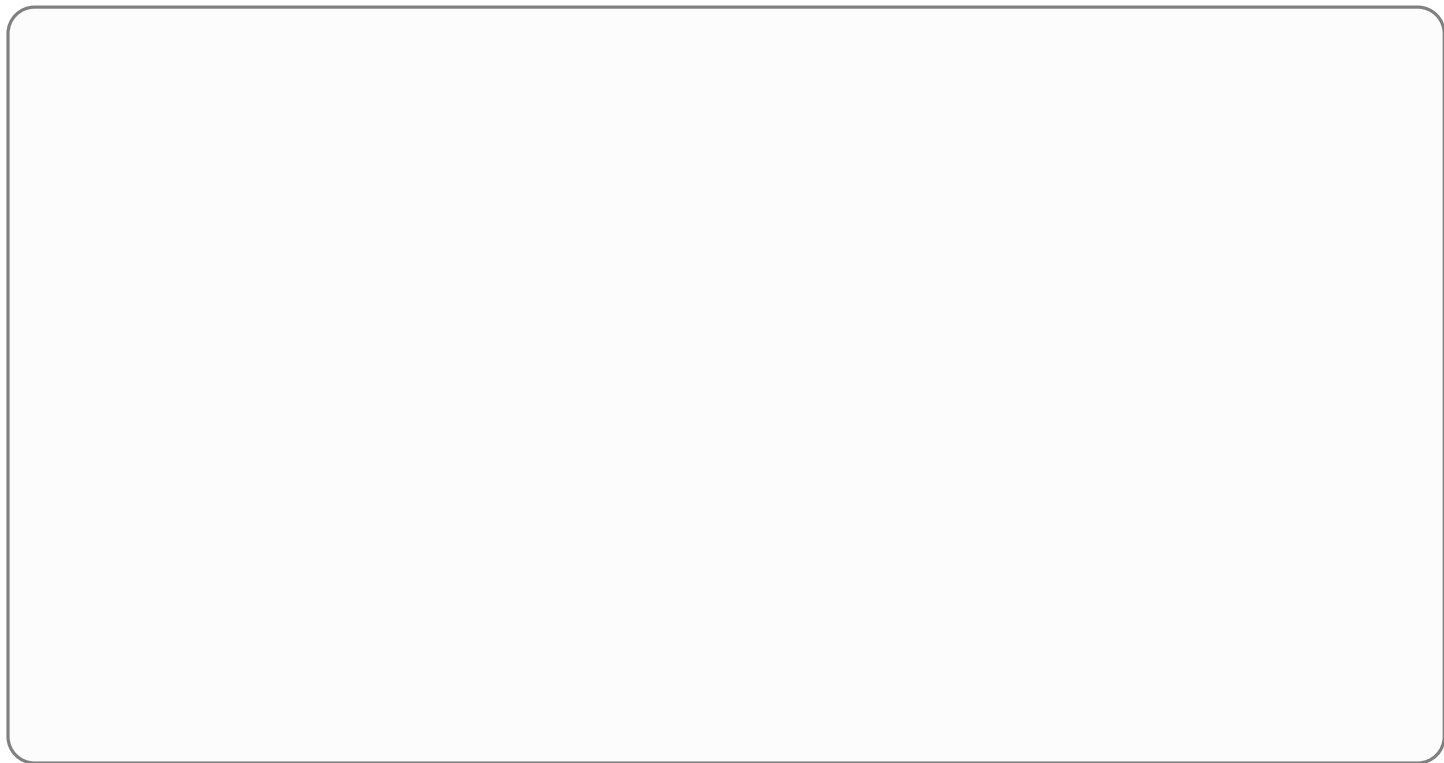
For each of those statements, say whether it is true, false, or uncertain (explain why and expose the necessary background).

1 – Children are like an inferior good.

2 – There was no growth on income during the Malthusian regime.

3 – When the probability at birth for life length L to be larger than ℓ is $P(L > \ell) = e^{-m\ell}$, then life expectancy at age 20 is equal to life expectancy at birth *minus* 20.

4 – Assume that the law of motion of a resource F (say a forest) is given by $\dot{F}(t) = rF(t) \left(1 - \frac{F(t)}{K}\right)$. In this case, there are only two steady state level of F .



5 – Having a high growth of money supply is a way to guarantee higher growth of real income in the long run.

